### 9.1.4 Karl Pearson's Correlation Coefficient

Karl Pearson, a great biometrician and statistician, suggested a mathematical method for measuring the magnitude of linear relationship between two variables say $X$ and $Y$. Karl Pearson's method is the most widely used method in practice and is known as Pearsonian Coefficient of Correlation. It is denoted by the symbol '$r$' and defined as

$$r = \frac{\text{cov}(X,Y)}{\sigma_x \sigma_y},$$

where $\text{cov}(X, Y) = \frac{1}{N}\sum(X-\overline{X})(Y-\overline{Y})$

$$\sigma_x = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(X_i-\overline{X})^2}$$

$$\sigma_y = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(Y_i-\overline{Y})^2}$$

Hence the formula to compute Karl Pearson Correlation coefficient is

$$r = \frac{\frac{1}{N}\sum_{i=1}^{N}(X_i-\overline{X})(Y_i-\overline{Y})}{\sqrt{\frac{1}{N}\sum_{i=1}^{N}(X_i-\overline{X})^2}\sqrt{\frac{1}{N}\sum_{i=1}^{N}(Y_i-\overline{Y})^2}}$$

$$r = \frac{\sum_{i=1}^{N}(X_i-\overline{X})(Y_i-\overline{Y})}{\sqrt{\sum_{i=1}^{N}(X_i-\overline{X})^2}\sqrt{\sum_{i=1}^{N}(Y_i-\overline{Y})^2}}$$

#### Interpretation of Correlation coefficient:

Coefficient of correlation lies between $-1$ and $+1$. Symbolically, $-1 \le r \le +1$

- When $r = +1$, then there is perfect positive correlation between the variables.
- When $r = -1$, then there is perfect negative correlation between the variables.
- When $r = 0$, then there is no relationship between the variables, that is the variables are uncorrelated.

Thus, the coefficient of correlation describes the magnitude and direction of correlation.

#### Methods of computing Correlation Coefficient

#### (i) When deviations are taken from Mean

Of all the several mathematical methods of measuring correlation, the Karl Pearson's method, popularly known as Pearsonian coefficient of correlation, is most widely used in practice.

$$r = \frac{\sum_{i=1}^{N}(X_i-\overline{X})(Y_i-\overline{Y})}{\sqrt{\sum_{i=1}^{N}(X_i-\overline{X})^2}\sqrt{\sum_{i=1}^{N}(Y_i-\overline{Y})^2}} = \frac{\sum xy}{\sqrt{\sum x^2 \sum y^2}}$$

Where $x = (X_i - \overline{X})$ and $y = (Y_i - \overline{Y})$;
$i = 1, 2 \ldots N$

This method is to be applied only when the deviations of items are taken from actual means.

#### Steps to solve the problems:

(i) Find out the mean of the two series that is $\overline{X}$ and $\overline{Y}$.

(ii) Take deviations of the two series from $\overline{X}$ and $\overline{Y}$ respectively and denoted by $x$ and $y$.

(iii) Square the deviations and get the total of the respective squares of deviation of $x$ and $y$ respectively and it is denoted by $\Sigma x^2$ and $\Sigma y^2$.

(iv) Multiply the deviations of $x$ and $y$ and get the total and it is denoted by $\Sigma xy$.

(v) Substitute the values of $\Sigma xy$, $\Sigma x^2$ and $\Sigma y^2$ in the above formula.



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